

A Comparative study of Support Vector Machines and Convolutional Neural Networks for Adult Income

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Abstract

This study investigates how well Support Vector Machines (SVM) and Convolutional Neural Networks (CNN) can predict whether an individual makes more or less than \$50,000 annually. The UCI Adult Income Dataset provides demographic, and employment information used in the research. Although it had difficulty correctly determining those with higher earnings, the SVM model was still effective. However, despite requiring more processing capacity, the CNN model was more effective in identifying higher incomes. For real-world jobs where accurately identifying lower and higher income groups is important, such as tax checks or loans, this makes CNN more beneficial. These results show which model may be more reliable for financial predictions, especially in cases where the data is not evenly distributed.

1. Introduction

Income prediction is an important data analysis topic that is relevant to business, social science, and economics. Policymakers, companies, and researchers can all benefit from understanding the factors that influence an individual's income. Extracted from the U.S. Census, the Adult Income dataset includes detailed data on employment, education, and demographic data. The aim of this study is to investigate this data and develop a predictive model that determines if a person's yearly income exceeds or falls below \$50,000 (Matkowski, 2021).

Support Vector Machines (SVM) and Convolutional Neural Networks (CNN) are the two methods examined in this study. SVM is frequently used to organise data by determining the best line for splitting different categories of income (Vapnik & Cortes, 1995). CNNs are often used to identify significant details or characteristics in different types of data (Lecun, et al., 2002). The study's evaluation of SVM and CNN provides useful knowledge into their specific performance, efficiency, and suitability for uses including income level sorting.

Preprocessing of the data is an important part of the study. The dataset consists of both numbers and categories; therefore, the categories must convert into a computer-understandable structure, deal with any missing values, and align the numbers. It's key to choose the appropriate data attributes because some may have a greater impact on income than others. A prepared dataset helps the models' efficient learning of important patterns. Two models - SVM and CNN are evaluated in the study. It aims to demonstrate the areas in which each model predicts income both effectively and poorly. The results may help select the best models for future tasks that are similar.

1.1. Support Vector Machines (SVM)

Machine learning uses Support Vector Machines (SVMs) frequently, especially in classification tasks. By selecting the most effective boundary, they separate multiple groups in data. They are good when group distinctions are obvious. Tasks like text sorting and image recognition utilise SVMs' ability to work well with data which has a lot of features and can prevent errors. But when dealing with big datasets, they could be slow and memory intensive. Making changes to SVM settings, such as choosing the appropriate kernel, can be challenging. However, despite the disadvantages, SVMs are a reliable choice for structured data that needs precise and clear category separation (Vapnik & Cortes, 1995).

1.2. Convolutional Neural Networks (CNN)

Machine learning models that are particularly good at observing patterns in photos are called Convolutional Neural Networks (CNNs). People don't need to provide them with important details like edges and shapes because they can learn these features on their own. For applications like face recognition, assisting self-driving cars, and analysing medical photos, CNNs are very useful. CNNs' high accuracy in handling difficult visual input is of its important advantages, especially when they are trained on large amounts of data. However, they require a lot of processing power and training takes a long time. Poor management might lead them to overfit, which means they perform well on the taught data but struggle with new images (Lecun, et al., 2002).

2. Dataset

The UCI Machine Learning Repository's Adult Income Dataset is used in this study. There are 48,842 rows in this dataset, including 14 demographic and job details. It attempts to sort people into two income brackets: those who make \$50,000 or less annually and those who make more than \$50,000. The dataset requires preprocessing because it contains numbers and labels. This includes steps such as encoding before to model training. The difference in income levels is an important feature of this dataset. Individuals who make \$50,000 or less dominate those who make above \$50,000. Table 1 provides an in-depth statistical summary to show this.

Table 1: Statistics Summary of the Dataset.

	Age	Educational-num	Capital-gain	Capital-loss	Hours-per-week	Income
Mean	38.64	10.08	1079.07	87.50	40.42	0.24
Standard Deviation	13.71	2.57	7452.02	403	12.39	0.43
Minimum	17	1	0	0	1	0
25%	28	9	0	0	40	0
50%	37	10	0	0	40	0
75%	48	12	0	0	45	0
Maximum	90	16	99999	4356	99	1

2.1. Initial Data Analysis

The graphs in figure 1 show the distribution of several features within the dataset. The largest groups on the age graph are in their 30s and 40s, with most people being between the ages of 20 and 60. The graphs for capital gain and loss show that few people have high values, however the majority have little to no profits or losses. This suggests that to improve the performance in the model, these features may need to be modified. Despite there is an even split in the education graph, a higher percentage of people had mid-level education, or between nine and twelve years. This provides support to the idea that income is strongly influenced by education.

The hours-per-week graph shows that, although some people work much more, most people work around 40 hours per week, which is average for full-time jobs. When estimating income, this data helps in separating between full-time and part-time employees. Finally, the income graph reveals that more than half of people make less than £50,000 than over. Although there isn't much of a difference, it's important to evaluate the model's performance using relevant metrics like precision and recall rather than just accuracy. To get the best results, these insights help us decide what factors to focus on and how to prepare the data.

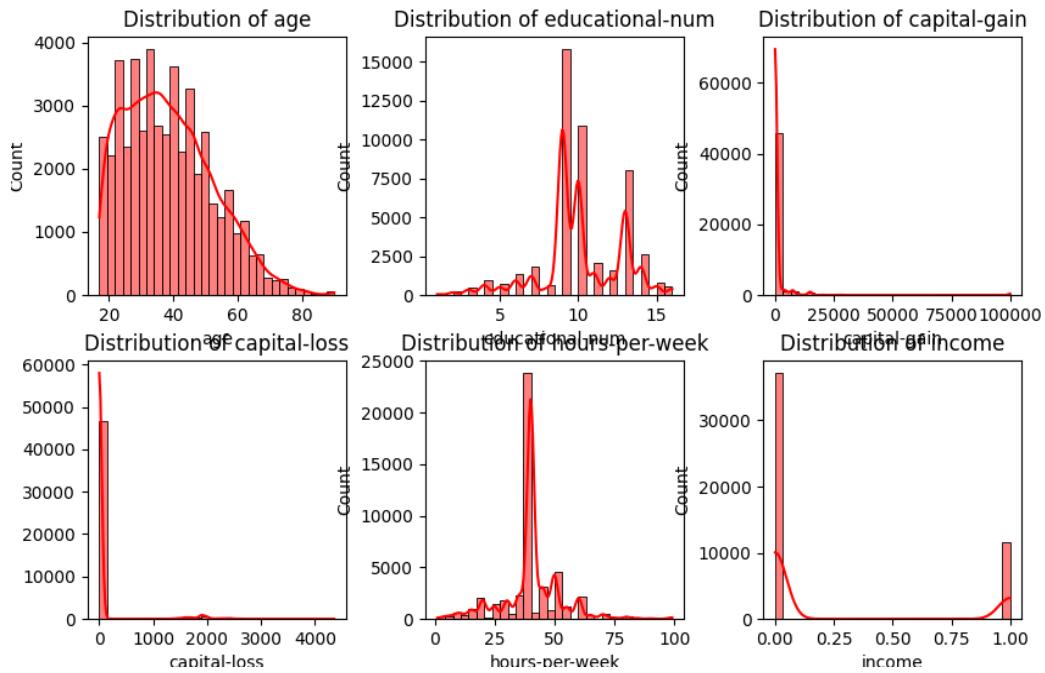


Figure 1: Histograms showing all the numerical features in the dataset.

3. Methodology

This study looks at how well a basic Convolutional Neural Network (CNN) and Support Vector Machines (SVM) predict income levels. Age, job, level of education, and other comparable personal characteristics are used to make predictions. Preparing and organising the data, creating and configuring the models, training the models on the data, and assessing how well each model predicts income levels are the key steps in the process.

3.1. Data Preprocessing

This study examines at how well a Convolutional Neural Network (CNN) and Support Vector Machines (SVM) predict income levels. It makes use of data regarding people's employment and past records. Preparing the data, setting up the models, training them, and assessing their performance are the main components of the study.

There are 48,842 rows and 14 characteristics in the Adult Income dataset. These characteristics are divided into categories and numbers. The technique known as LabelEncoder, was used for the category characteristics. Word-based categories are converted into numerical values using this technique (Alkhamash, et al., 2022). By doing this, the data becomes appropriate for model training.

Target labels (Y) and input features (X) were the two groups into which the data was separated. The data was then divided into two sections: 20% was kept for testing and the remaining 80% was used for training. To ensure that every feature was measured on the same scale, a program called StandardScaler was used (Raju, et al., 2020). When using neural networks, this step helps maintain the stability of the training process and is important for certain algorithms, such as SVM.

3.2. Parameters used for Support Vector Machine (SVM)

For tasks with two outcomes, a linear kernel SVM classifier was used. To ensure consistency, the classifier was trained on data that had been reduced to the same value. It then predicted various test data. Measures of accuracy, precision, recall, and F1-score were used to assess the model's performance. A confusion matrix was also produced, which gave an overview of predictions that were both correct and incorrect.

3.3. Parameters used for Convolutional Neural Network (CNN)

CNNs are usually used for tasks involving images, this study adapts a feedforward CNN to work with structured tabular data. Two completely connected hidden layers with ReLU activation are part of the model. The model can do binary classification because of an output layer with SoftMax activation that comes after these layers (Chai, et al., 2021).

The dataset was set up using DataLoader objects and transformed into PyTorch tensors to enable mini-batch training. Cross-entropy loss and the Adam optimiser were used to train the model across ten epochs. The learning progress was evaluated and looked for indications of underfitting or overfitting by tracking training and test loss at each epoch.

Predictions for the test set were produced after training. Like the SVM, classification metrics and a confusion matrix were used to assess the CNN's performance. To determine whether model performed better on this categorisation job, the results were directly compared.

3.4. Evaluation Metrics

Both approaches were evaluated using the same set of assessment metrics to ensure a fair comparison between the two models. The percentage of accurate predictions was measured using accuracy. For assessing performance on the neglected income class, precision and recall were important since they show that the model can detect real positives without overpredicting. In the context of binary classification, the F1-score offered an improved evaluation of performance by finding a balance between precision and recall. Additionally, each model's ability to separate between the two income groups was visualised using a confusion matrix. Training and validation loss were plotted across epochs to analyse the CNN's training behaviour. This helped in evaluating its suitability for managing structured, tabular data and better understanding its learning dynamics.

4. Results, Findings & Evaluation

4.1. Model Results and Performance Comparison

Table 2's results show that the model has problems with class imbalance. Overall, it achieves an accuracy of 82%; however, the two classes' recall scores differ significantly. Recall for class 1 is only 0.30, compared to 0.97 for class 0. The smaller class achieves only 0.89 while the larger class scores 0.44 F1-scores, which shows the difference in scores. Regardless of the number of samples in each class, the macro average metrics, which provide a recall of 0.64 and an F1-score of 0.66, indicate this problem because they assess both classes equally.

Table 3 shows significant progress in key areas. With a recall score of 0.93, the CNN does well in the majority class. Also, it raises the recall for the minority class of the SVM to 0.62, nearly doubling it. As a result, the overall accuracy rises to 86%. At 0.91 and 0.67, the F1-scores for both categories are also much higher. Recall and F1-score have weighted averages of 0.86 and 0.85, respectively. This demonstrates that CNN is far more effective at managing the irregular dataset while maintaining prediction accuracy.

Table 2: SVM Performance Metrics.

SVM				
	Precision	Recall	F1-score	Support
0	0.82	0.97	0.89	7479
1	0.79	0.30	0.44	2290
Accuracy			0.82	9769
Macro Average	0.80	0.64	0.66	9769
Weighted Average	0.81	0.82	0.78	9769

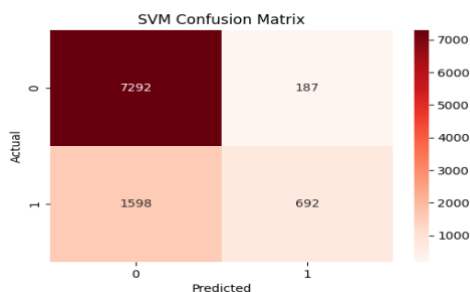
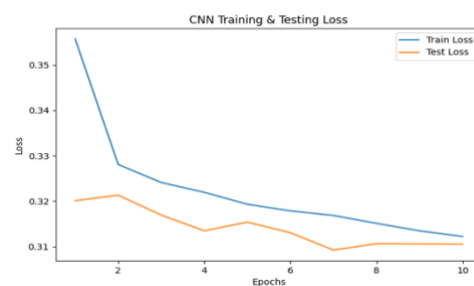
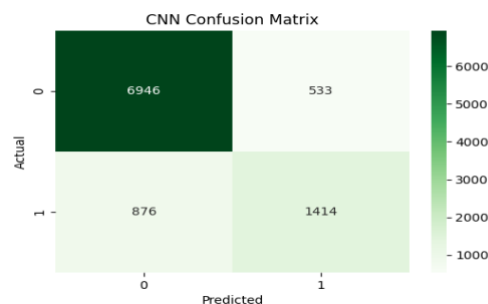
Table 3: CNN Performance Metrics.

CNN				
	Precision	Recall	F1-score	Support
0	0.89	0.93	0.91	7479
1	0.73	0.62	0.67	2290
Accuracy			0.86	9769
Macro Average	0.81	0.77	0.79	9769
Weighted Average	0.85	0.86	0.85	9769

4.2. Model Behaviour

CNN's learning process during training is shown in Figure 3. Training loss is shown by a blue line on the graph, while test loss is shown by the orange line. Both lines exhibit downward progress, indicating successful learning. This indicates that the model learnt correctly without memorising the training data, which is known as overfitting and would result in poor results on new data. The CNN works well on both seen and unseen data, shown by the small space between the lines. By the final epoch, the lines reached a loss value of about 0.31. This suggests that the model gathered as much information as possible from the data available.

The different prediction patterns of figure 3 can be observed when comparing them. The findings in the confusion matrix of the SVM (Figure 2) are not distributed. It performed well at predicting 7,292 low-income cases, but it had difficulties with high-income cases where only 692 cases were predicted correctly. Many errors were made, resulting in 1,598 false negatives, where high salaries were mistakenly classified as low. The CNN matrix in Figure 4, however, is more evenly distributed. It also performs well in low-income cases, identifying 6,946 (less than SVM), and performs significantly better in high-income predictions, correctly identifying 1,445 and making fewer significant errors.

**Figure 2: Confusion matrix for SVM.****Figure 3: Training and Testing Loss graphs for CNN.****Figure 4: Confusion matrix for CNN.**

5. Conclusion

This study looked at two ways for predicting an individual's income based on personal characteristics like age and education. About 82% of the time, earnings were accurately predicted using the first method, SVM. However, only around 30% of those making above \$50,000 were accurately identified, and it regularly failed to identify such individuals. The success rate of CNN, the second approach, was higher. 86% of the time, it was accurate, and 62% of the time, it identified top earners. In real-world scenarios where it's critical to accurately determine both lower and higher income levels, CNN becomes more useful as a result.

CNN learnt slowly and avoided errors, but it required more processing power to train. It was improving during the training procedure, as shown by the graphs. In comparison, the SVM used shortcuts but operated quicker. Often, it would estimate lower incomes because those predictions were simpler and produced good outcomes. As can be seen from the visuals, this method explains why it often ignored those with larger salaries.

These figures highlight the practical applications of CNN. For example, governments studying tax data or banks analysing loan applications may make costly errors if they fail to recognise high incomes, as the SVM often does. Because it works more consistently, as shown by its matrix, the CNN is more reliable and safer for important decisions. As demonstrated by its progressive learning curve in Figure 2, training a CNN uses more computer resources, but the results tend to be better (Lu, et al., 2017).

Given that the losses were still slightly declining at the end, the training graph suggests that the CNN might perform better with additional training time. It would be useful to test this in the future, even though it isn't obvious in the current results. Investigating further training could result in improved performance and a greater understanding of CNN's capabilities.

6. References

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